# RAJAVELAVAN APPAIYACHETTY

Junior Software Developer

LinkedIn | GitHub | appaiya.raja@gmail.com | 7904127379 | Chennai, Salem (Open to relocate)

## SUMMARY

Full Stack Software Engineer with 2 years of experience building scalable web applications using React.js, Node.js, AWS, and REST APIs. Currently building AI-powered applications with Python, LangChain, LangGraph, and LLMs, with hands-on experience in Agentic AI and Retrieval-Augmented Generation (RAG). Passionate about developing secure, production-ready AI solutions.

## SKILLS

**AI & LLM** — LangGraph, LangChain, Agentic AI, RAG, Prompt Engineering, Vector Databases (pgVector), OpenAI API, Llama 3, Embeddings, Tokenization
**Backend** — Node.js, Express.js, NestJS, Python, FastAPI, REST APIs, JWT Authentication, RBAC
**Frontend** — React.js, Next.js, Angular, JavaScript (ES6+), TypeScript, HTML5, CSS3, Tailwind CSS, Redux Toolkit
**Database** — PostgreSQL, MySQL, MongoDB
**Cloud & DevOps** — AWS (EC2, S3, IAM, VPC, CloudWatch), Docker, Kubernetes, Terraform, Jenkins, GitHub Actions
**Dev Tools** — Git, GitHub, Linux, Bash, Postman

## PROFESSIONAL EXPERIENCE

**Junior Software Developer, Code Flakes Studio** 02/2025 – Present | Chennai
*Tech Stack: React JS · Angular · Node.js · NestJS · MySQL · Knex.js · AWS*

- Engineered end-to-end product features using React JS and Node.js/Express — delivering responsive UI components and secure REST APIs from a single requirement, enabling consistent on-schedule sprint releases.
- Reduced feature delivery time by building a reusable library of React components and shared utility hooks, eliminating duplicate code and improving UI consistency across the product.
- Achieved a 30% improvement in application performance by profiling and optimising MongoDB query execution and React rendering pipeline through targeted indexing, lazy loading, and backend caching strategies.
- Secured multi-role applications by implementing JWT authentication and role-based access control (RBAC) with Express middleware, enforcing principle-of-least-privilege data access across all user tiers.
- Delivered production-ready releases through active participation in Agile sprints, code reviews, and cross-functional collaboration with designers, QA engineers, and stakeholders; deployed applications on AWS EC2 and S3.

## PROJECTS

**Enterprise AI Security Gateway** | *FastAPI · OpenTelemetry · Httpx · Pydantic · RedisVL*

- Built a high-performance AI Security Gateway (FastAPI/Python) featuring a custom Data Loss Prevention (DLP) engine that utilizes the Aho-Corasick algorithm for highly efficient, real-time redaction of outbound API requests.
- Secured enterprise LLM adoption by deploying a centralized "AI Firewall" that prevents "Shadow AI" data leaks, ensuring sensitive PII and corporate secrets never leave the internal network.
- Driving the architectural roadmap to implement vector-based Semantic Caching (RedisVL) to reduce LLM inference costs by up to 80%, alongside zero-knowledge cryptographic auditing for tamper-proof compliance trails.

**DevSecOps AI Engine** | *FastAPI · LangGraph · Gemini LLM · Ollama · pgvector · AWS*

- Architected an autonomous DevSecOps AI engine (FastAPI, LangGraph) to analyze AWS security alerts and dynamically generate remediation scripts for human-in-the-loop approval.
- Developed a React dashboard featuring real-time AI reasoning streams (SSE) and a PostgreSQL/pgvector database for semantic memory, similarity search, and tamper-proof audit logging.
- Designing the production rollout by containerizing the application with Docker, configuring GitHub Actions for CI/CD, and provisioning AWS infrastructure via Terraform.

**Handloom Materials Management System** | *React.js · Redux Toolkit · Node.js · Express.js · MongoDB · JWT · Tailwind CSS — Live Demo*

- Built and deployed a full-stack MERN application for inventory, purchase, sales, order, and customer management with secure role-based access control.
- Developed RESTful APIs, JWT authentication, OTP-based email verification, and automated invoice generation to streamline business workflows.
- Implemented Redux Toolkit for centralized state management and designed real-time dashboards for inventory tracking and order lifecycle monitoring.

## EDUCATION

**Master of Computer Applications (MCA), TPGIT** 10/2022 – 08/2024 | Vellore, TN
*CGPA: 8.0 / 10*

**Bachelor of Computer Applications (BCA), KSRCAS** 06/2019 – 06/2022 | Tiruchengode, TN
*CGPA: 7.8 / 10*

## CERTIFICATES

- Data Structures and Algorithms in JavaScript - FreeCodeCamp
- AWS Cloud Computing Certification - ACTE DevOps course certification
- AWS Technical Essentials - AWS